

PROJECT PORTFOLIO

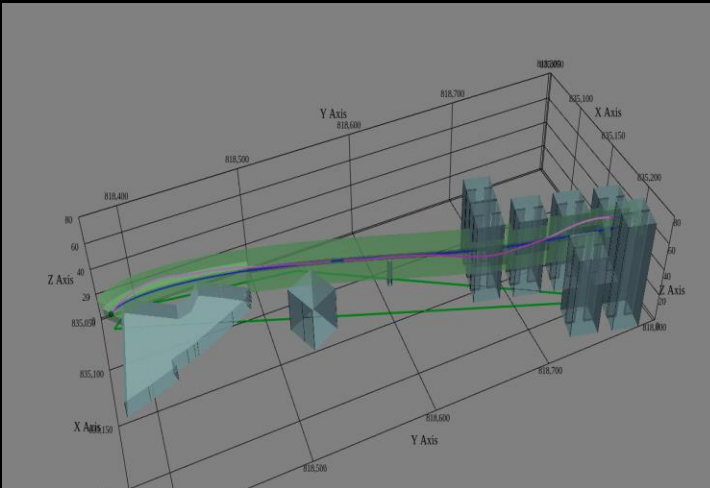
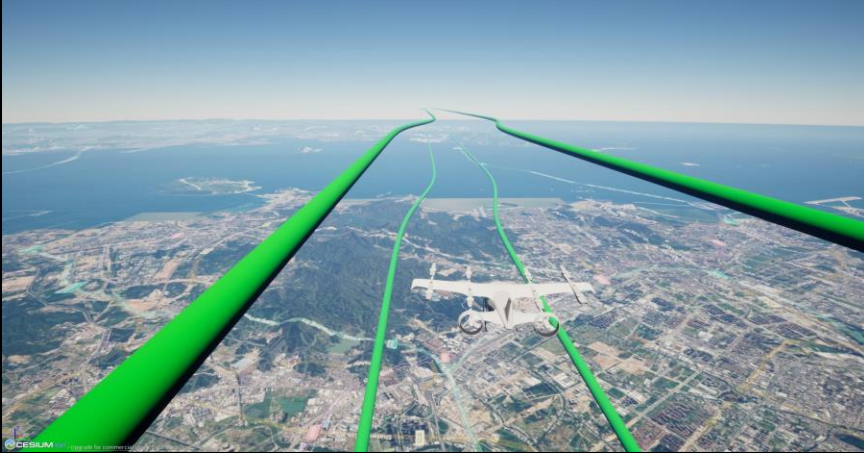
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ABOUT ME



👋 Hi, I am Dave Chan, a Research Associate in the Department of Computing at The Hong Kong Polytechnic University (Hong Kong SAR). I completed my PhD in Aeronautical and Aviation Engineering at The Hong Kong Polytechnic University in 2025 under the supervision of Professor Kam Hung Ng, and I hold a B.Eng. in Mechanical Engineering from The University of Hong Kong (2015). My research program centers on system dynamics, nonlinear optimal control, trajectory optimization, and machine learning, with an emphasis on developing methods and algorithms for autonomous aerial vehicles.

You can find more about me in my CV. Please get in touch. I am always looking for new collaborations 😊

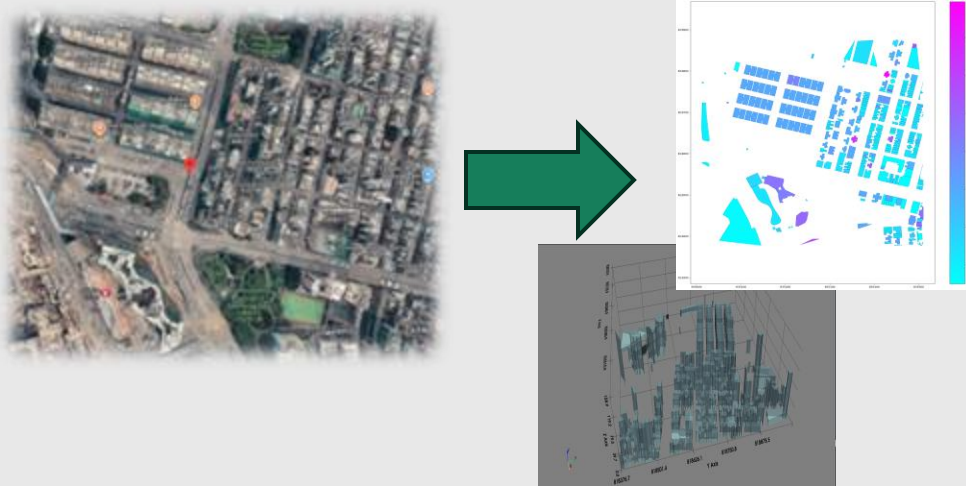
SELECTED PHD RESEARCH HIGHLIGHTS



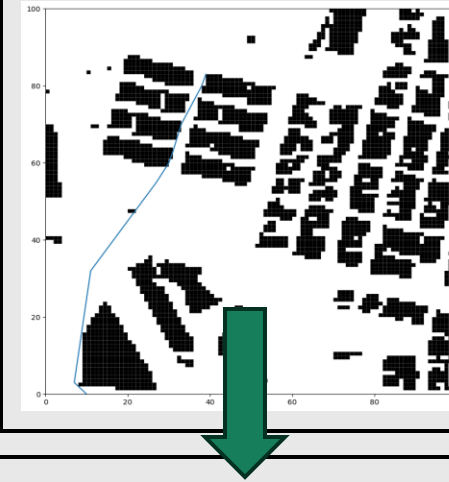
UAV TIME-OPTIMAL MOTION PLANNING FRAMEWORK USING NOVEL CURVE FRAMING METHOD

Urban Environment Modeling:

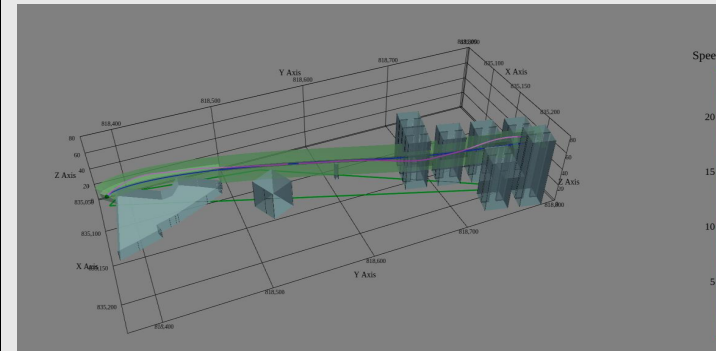
- Readily available building and structure data (HK Lands Dept / OpenStreetMap etc.)



Convert to a grid map and compute a global path (predefined or computed in real time):



Generating the initial flight corridor:



The position and velocity of any vehicle can be reformulated using a **Gravity Normal frame**

$$s^*(t) = \min_{s \in [s_0, s_f]} \|p_Q(t) - \Gamma(s)\|^2 \quad \dot{s}^*(t) = \frac{\mathbf{e}_1(s) \cdot \mathbf{v}_Q(t)}{\nu(s) - \lambda_3(s)w_1(t) + \lambda_2(s)w_2(t)}$$

$$w_1 = \mathbf{e}_2^T(s^*) ({}^I \mathbf{p}_Q - \Gamma(s^*)) \quad \dot{w}_1(t) = \mathbf{e}_2(s) \cdot \mathbf{v}_Q(t) + \dot{s}(t)\lambda_1(s)w_2(t)$$

$$w_2 = \mathbf{e}_3^T(s^*) ({}^I \mathbf{p}_Q - \Gamma(s^*)) \quad \dot{w}_2(t) = \mathbf{e}_3(s) \cdot \mathbf{v}_Q(t) - \dot{s}(t)\lambda_1(s)w_1(t)$$

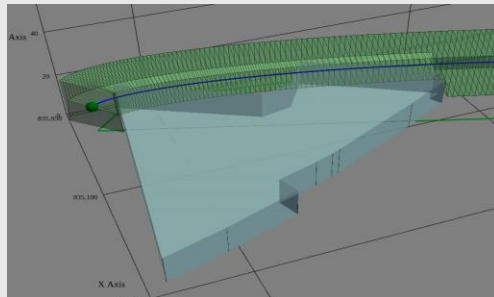
Chan, Y. Y., Shu Y. X., Chen, W. H. & Ng, K. K. H. (2026). Time-Optimal Motion Planning of Cable-Suspended Payload Quad-copters Using Spatial Reformulation. Aerospace Science and Technology. (2026 SCI IF: 5.8, JCR: Q1).

Chan, Y. Y., , Ng, K. K. H., Wang, T., Hon, K. K., & Liu, C. H. (2025). Near time-optimal trajectory optimisation for drones in last-mile delivery using spatial reformulation approach. Transportation Research Part C: Emerging Technologies, 171, 104986. (2026 SCI IF: 7.6, JCR: Q1)

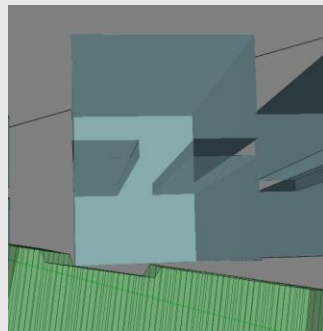
UAV TIME-OPTIMAL MOTION PLANNING FRAMEWORK USING NOVEL CURVE FRAMING METHOD

Further adjustments to the corridor:

- Designed an algorithmic routine to compute the final safe corridor.
- Eliminating intrusions



Vertical (overhead) intrusion into the corridor



Lateral (side) intrusion into the corridor

Direct Multiple Shooting for Global Time-Optimal Trajectory:

$$\min_{x_\xi(\cdot), u(\cdot), T} \quad s.t.$$

$$x_\xi(0) = x_0, \quad x_\xi(T) = x_T,$$

$$\dot{x}_\xi(t) = f(x_{\xi,k}(t), u_k(t)),$$

$$x_{\xi,k}(t) \leq x_{\xi,k}(t) \leq \overline{x_{\xi,k}(t)},$$

$$u_k(t) \leq u_k(t) \leq \overline{u_k(t)},$$

Cost Function: flight time T

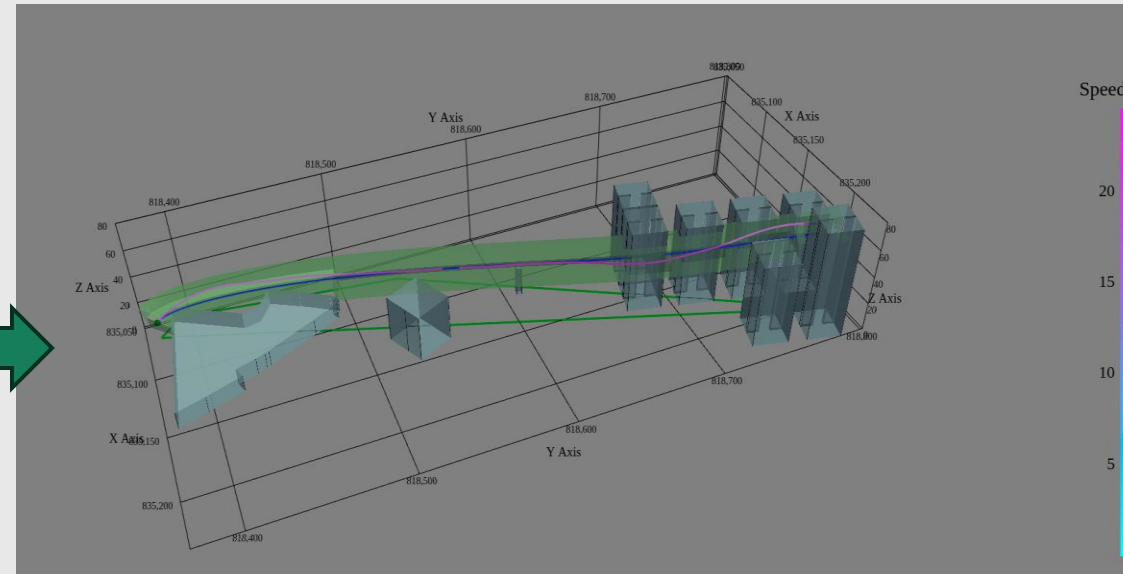
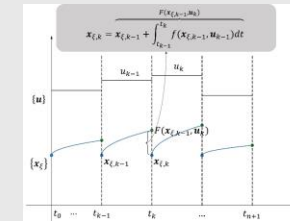
Decision variables: state and control trajectory

Initial and terminal state constraints.

Dynamic constraints

State bounds (governed by the corridor)

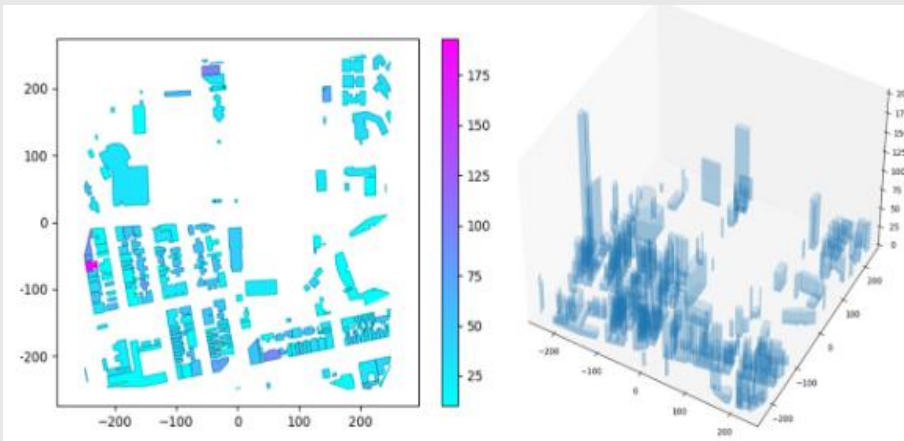
Control bounds



Chan, Y. Y., Shu Y. X., Chen, W. H. & Ng, K. K. H. (2026). Time-Optimal Motion Planning of Cable-Suspended Payload Quad- copters Using Spatial Reformulation. Aerospace Science and Technology. (2026 SCI IF: 5.8, JCR: Q1).

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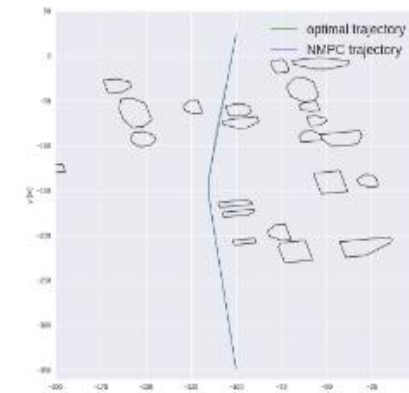
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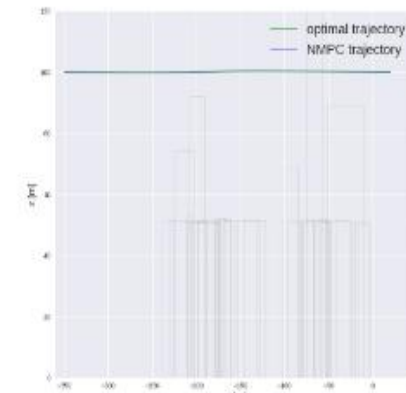
Expected time for Time-Optimal Trajectory			
OD Pair	Distance (m)	Flying Time (s)	Compute Time (s)
1	255.31	15.80	13.55
2	287.93	18.77	13.78

NMPC Tracking results			
OD Pair	Distance (m)	Flying Time (s)	Corridor Gen. Time (ms)
1	260.70	16.20	0.41
2	298.98	19.05	0.43

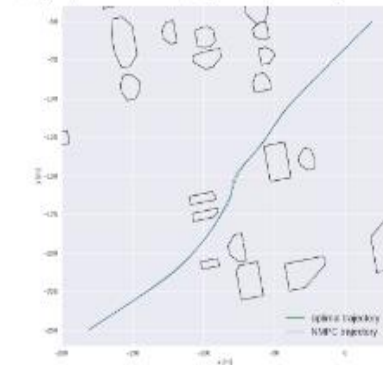
- The offline trajectory module assumes an ideal, instantaneous controller response, which contributes to the discrepancy between planned and actual performance.



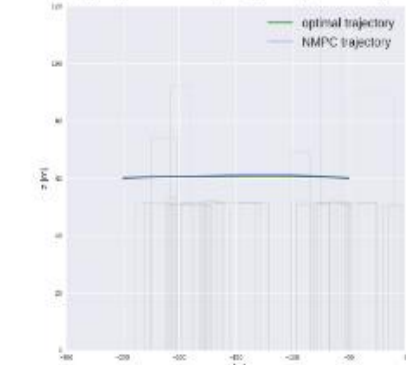
(a) projection on x-y plane for OD pair 1



(b) projection on y-z plane for OD pair 1



(c) projection on x-y plane for OD pair 2

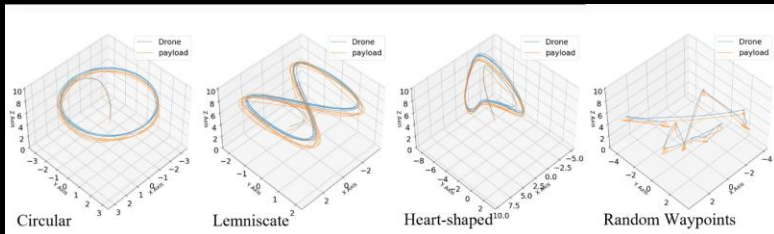


(d) projection on y-z plane for OD pair 2

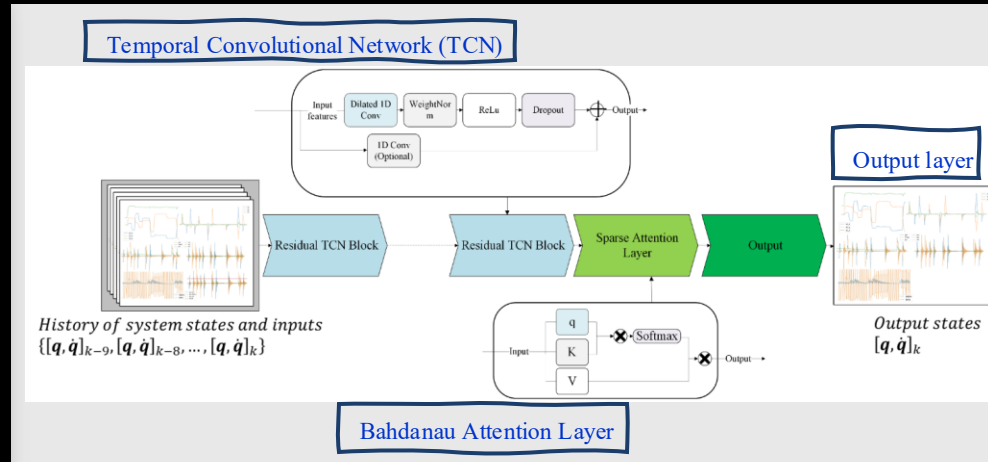
Chan, Y. Y., Shu Y. X., Chen, W. H. & Ng, K. K. H. (2026). *Time-Optimal Motion Planning of Cable-Suspended Payload Quad- copters Using Spatial Reformulation.* *Aerospace Science and Technology.* (2026 SCI IF: 5.8, JCR: Q1).

Chan, Y. Y., , Ng, K. K. H., Wang, T., Hon, K. K., & Liu, C. H. (2025). *Near time-optimal trajectory optimisation for drones in last-mile delivery using spatial reformulation approach.* *Transportation Research Part C: Emerging Technologies,* 171, 104986. (2026 SCI IF: 7.6, JCR: Q1)

PHYSICS-INFORMED ATTENTION TEMPORAL CONVOLUTIONAL NETWORK (PIATCN) FOR MODELLING OF CABLE SUSPENDED QUADCOPTERS



Simulated flights were conducted in the PX4 SITL platform.



Designed to capture complex nonlinear behaviors while ensuring physically consistent predictions.

Uses physics-informed constraints to reduce the required training data

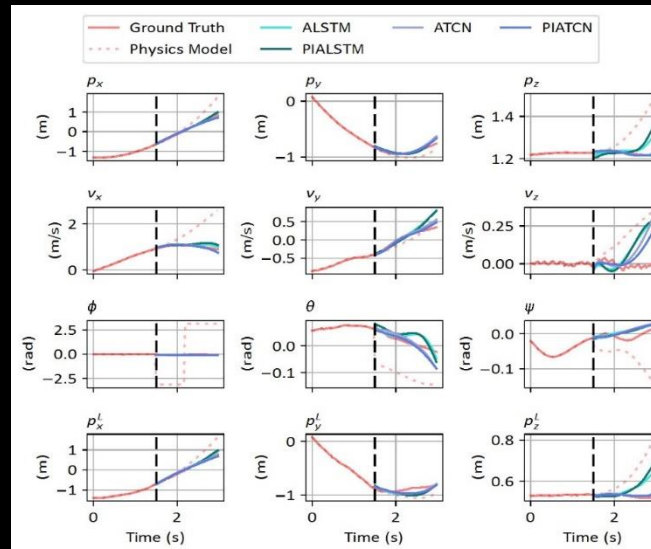


Table 3.1: Comparison of PIATCN with all the baselines

	Physics model	ALSTM	PIALSTM	ATCN	PIATCN
Position [m]	0.458	0.531	0.427	0.250	0.151
Velocity [m]	0.924	0.415	0.207	0.423	0.077
Orientation [rad]	0.245	0.047	0.083	0.100	0.050
Payload Position [m]	0.424	0.953	0.490	0.346	0.319
Combined	2.051	1.946	1.207	1.119	0.597
Average Inference [ms]					
I40	0.020	18.37	19.23	17.14	19.62
Nvidia Jetson Nano	2.14	110.55	114.51	119.46	127.85

Outperforms LSTM, ALSTM and baseline TCN models.

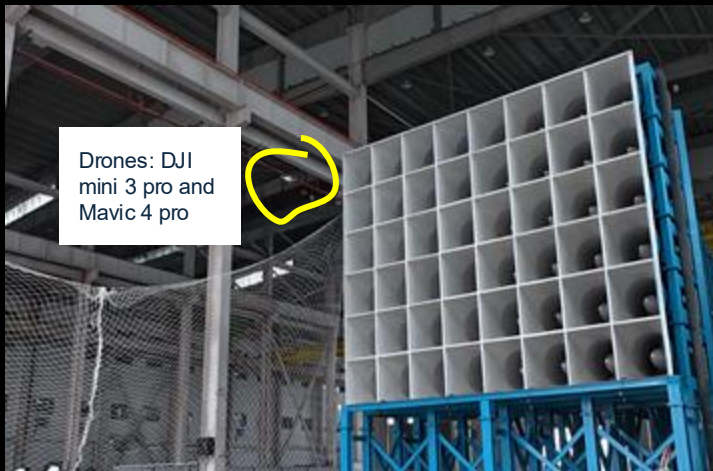
Chan, Y. Y., Ng, K. K. H., Liu, Y. (2025). Dynamic Modeling of Cable-Suspended Payload Quadcopter: A Physics-Informed Deep Learning Approach. In AIAA AVIATION FORUM AND ASCEND 2025

SELECTED PROJECT HIGHLIGHTS

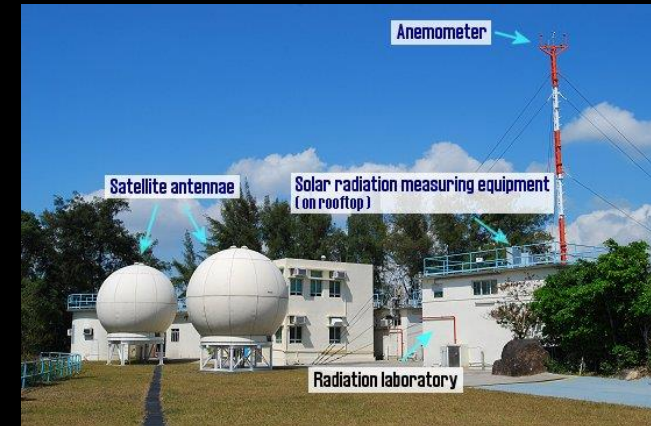


RESEARCH ASSOCIATE

THE DEPARTMENT OF COMPUTING, HONG KONG POLYTECHNIC UNIVERSITY



Wind wall tests using commercial-grade drones in GARA — to validate the accuracy of third-party wind-estimation platforms (e.g., AirData).



Flight test in King's Park Meteorological Station and Siu Ho Wan Wind automatic weather station (for validation)

Drone-based wind-speed and direction estimation using onboard flight data, with both machine learning and system-dynamics approaches (**ongoing**).

Collaborated closely with the Hong Kong Observatory (**HKO**) and the Guangdong Aerospace Research Academy (**GARA**).

RESEARCH ASSOCIATE

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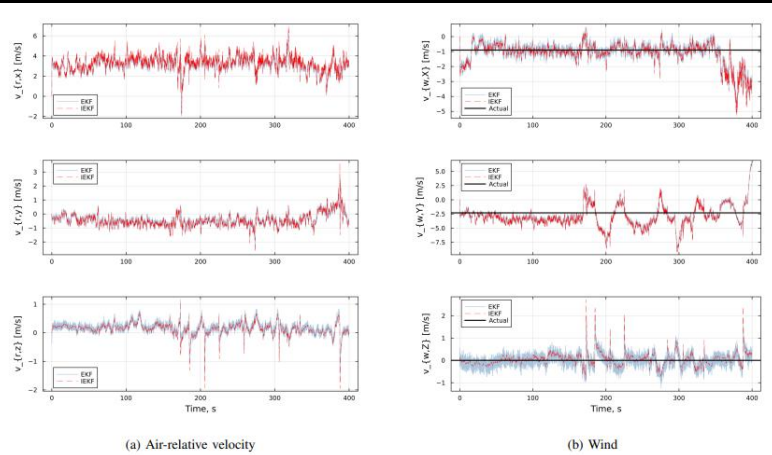


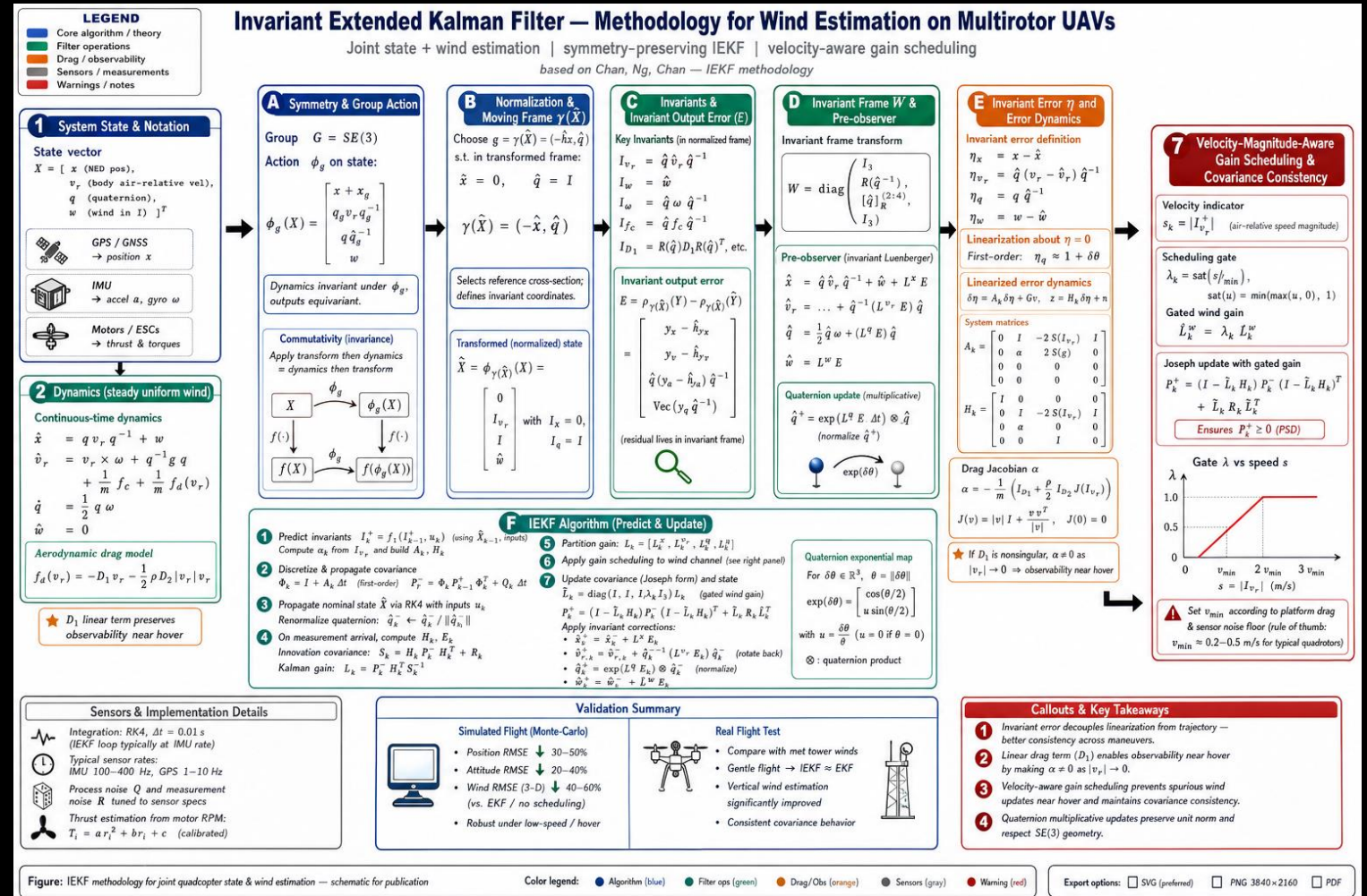
Fig. 6: Comparison of the IEKF and the EKF wind estimation for the real flight: (a) air-relative velocity, (b) wind velocity.

A Kalman filter-based approach is also adopted.

The proposed IEKF can estimate wind speed and direction in real flight tests



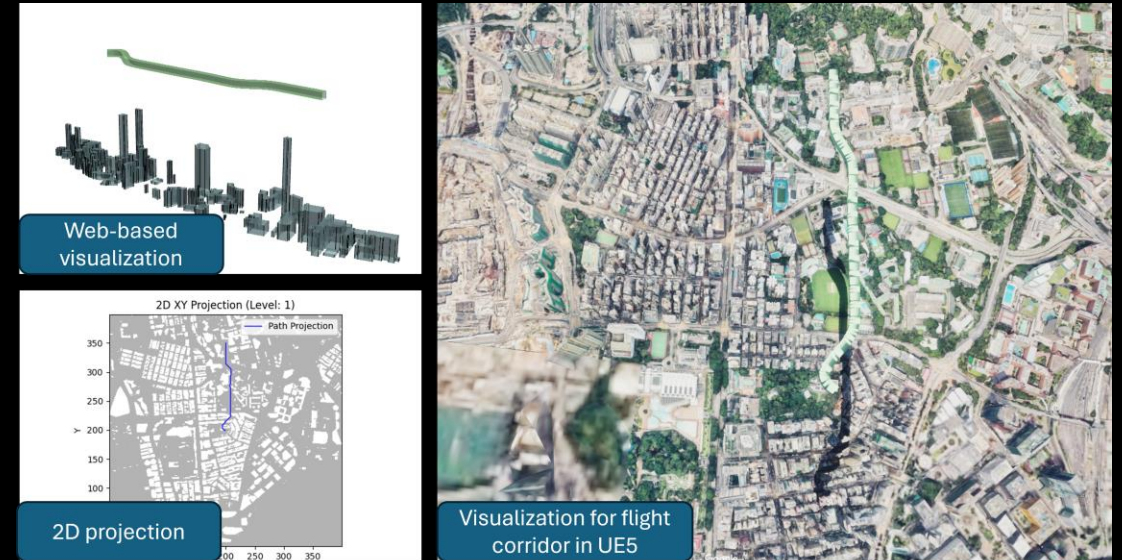
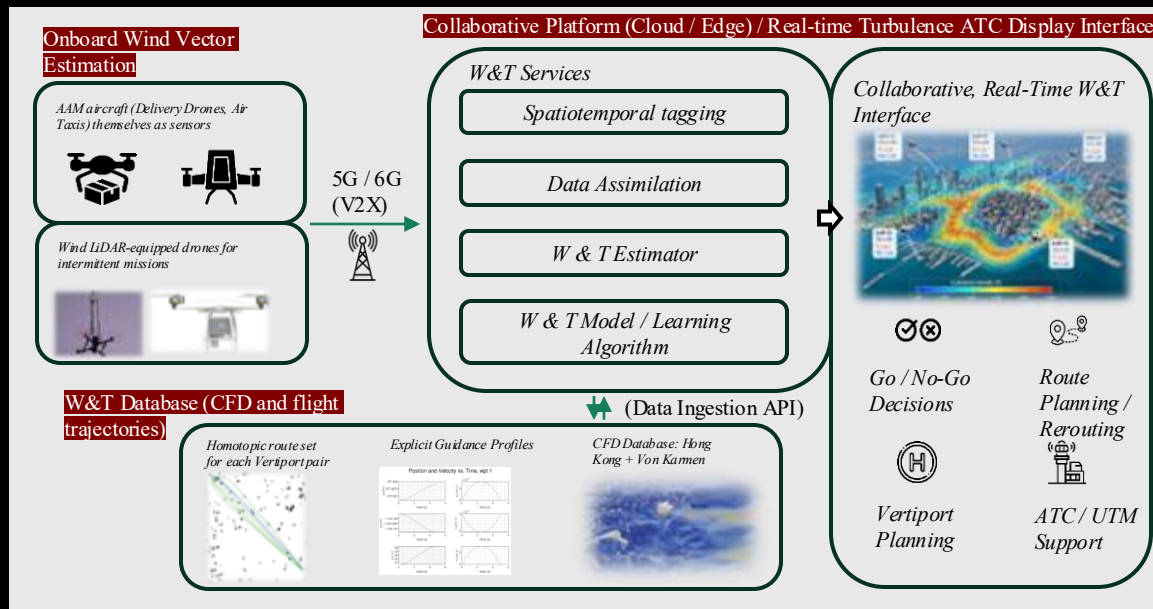
Improving the IEKF based on the symmetry-preserving properties of the dynamical system



Chan, Y. Y., Ng, K. K. H. & Pak Wai Chan (2026). Invariant Extended Kalman Filtering for Wind Estimation on Commercial Multirotor UAVs. Submitted to IEEE Transactions on Intelligent Transportation Systems (T-ITS). Under first review.

RESEARCH ASSOCIATE

THE DEPARTMENT OF COMPUTING, HONG KONG POLYTECHNIC UNIVERSITY



Cosys-AirSim + UE5 to model realistic UAV flight dynamics and sensor suites (GNSS/IMU/cameras), terrain, vegetation and buildings.

Collaborated closely with the **Hong Kong Observatory (HKO)**.

Multi-sensor integration mapping to build real-time Wind and Turbulence (W&T) maps for ATC operation and guidance (**Ongoing**).

ROBOTICS ENGINEER

RV AUTOMATION TECHNOLOGY COMPANY LIMITED



Smart Forklift project - cooperation with Linde Material Handling HK Ltd.



Smart Wheelchair



Smart sanitation robot



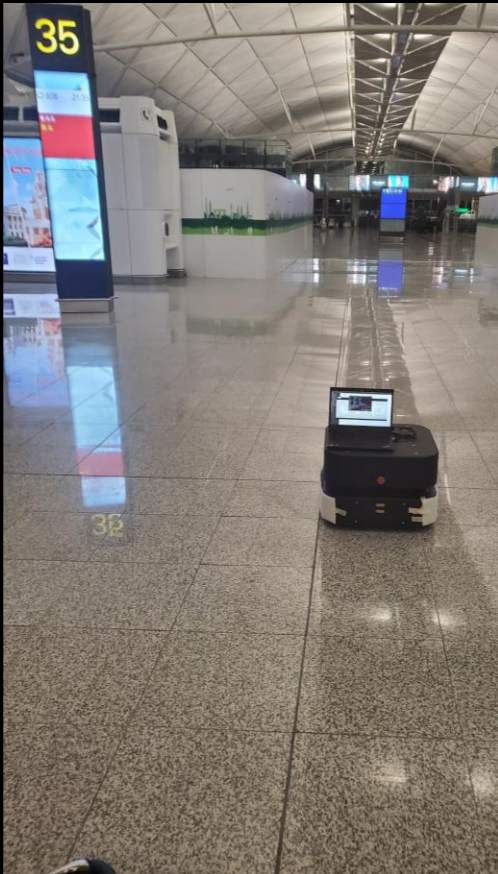
Developed various types of Autonomous Mobile Robots (AMRs), including logistics robots, smart forklifts, smart wheelchairs, etc.

Implemented and tuned system designs, motion-planning algorithms, and SLAM algorithms using C++ and Python on Robot Operating System (ROS) platforms

Patent: 周定汉, 陈彦元, 刘显杰, 陈炎材, 李逸曦, 伍淦鸿, "Robot Cooperation System Composed of a Picking Robot and a Delivery Robot", Chinese Utility Model Patent CN216913853U, Jul. 2022

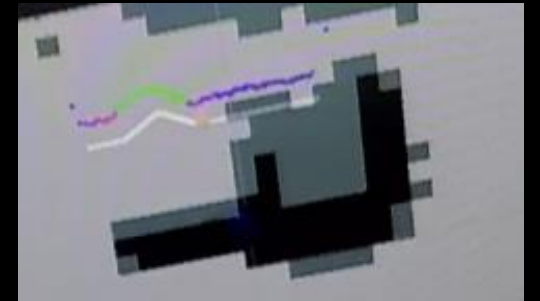
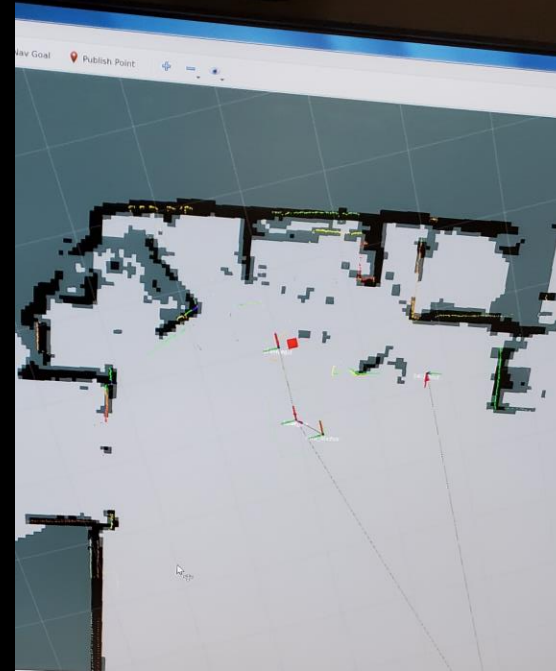
ROBOTICS ENGINEER

RV AUTOMATION TECHNOLOGY COMPANY LIMITED



Implemented our global and local planners

Implemented our SLAM algorithm (graph-optimization based).

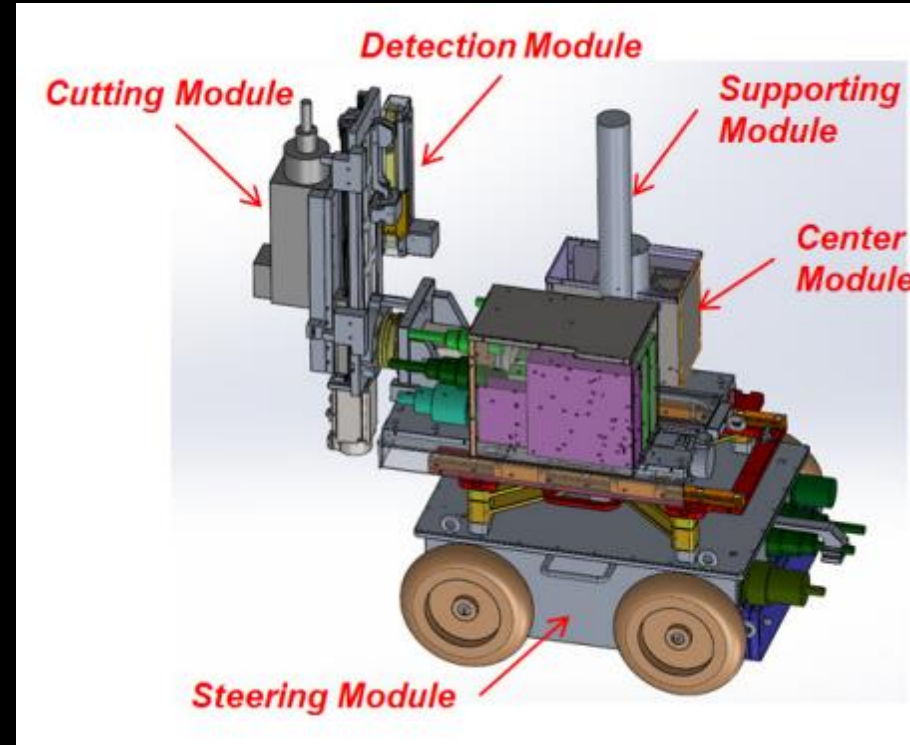
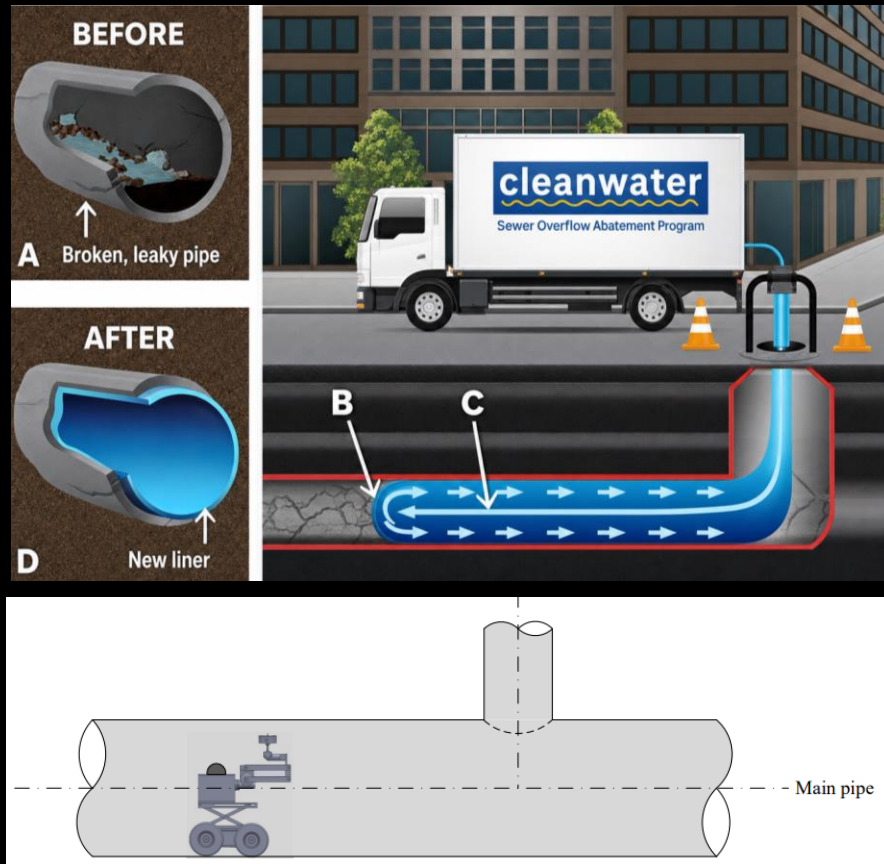


Localization-independent auto-docking.

Developed various types of Autonomous Mobile Robots (AMRs), including logistics robots, smart forklifts, smart wheelchairs, etc.

Implemented and tuned system designs, motion-planning algorithms, and SLAM algorithms using C++ and Python on Robot Operating System (ROS) platforms

RESEARCH ASSOCIATE, HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY



Cured-in-place pipe (CIPP) rehabilitation — repair robot cutter.

Developed an automatic robot cutter that can cut the lateral connection branch opening during underground drainage-pipe repair in Hong Kong.

THANK YOU

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